

ECONOMICS

Paper 3 Multiple Choice

9708/32

October/November 2017

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.



Done!

Ang

Larak mushtaq

Solved with reasons!

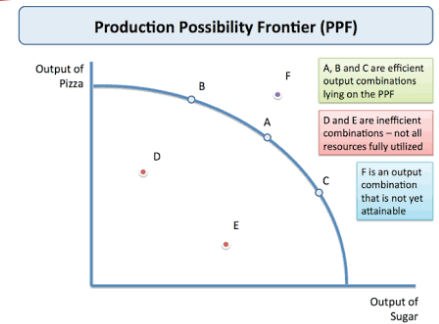
1hr 15mins for '30 MCQs'

This document consists of **10** printed pages and **2** blank pages.

- 1 In an economy no one can be made better off without making someone else worse off.

What can be deduced from this?

- (Newer)
- A Individuals are the best judges of their own well-being.
- B Individuals can be relied upon to behave rationally. (They don't)
- C The distribution of income is socially optimal. (It's never)
- ☒ D The economy's resources are allocated efficiently. (Yes)



- 2 In 2015, a large mining company said it would reduce the number of staff by 6000 and sell its less profitable mines in an attempt to become more efficient.

If it is successful, what is most likely to happen to its costs?

	total fixed cost	total variable cost	average cost
☒ A	fall ✓	fall ✓	fall ✓
B	fall ✓	fall ✓	no change
C	fall ✓	no change	fall ✓
D	no change	fall ✓	fall ✓

* Staff reduction :- Variable cost ↓

* selling its less profitable mines :- Fixed cost ↓

* Becoming more efficient :- Average cost ↓

- 3 The production of a firm which operates in an imperfectly competitive market gives rise to external production costs.

Which statement about this firm must be correct?

A External costs exceed external benefits.

B Private costs exceed social costs. (can't happen)

☒ C Social costs exceed private costs.

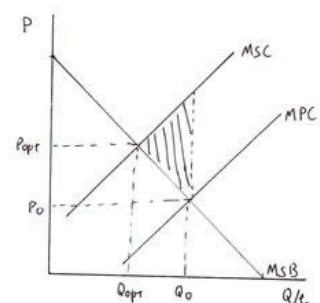
D Social costs exceed social benefits. (still, EB can be greater than EC)

$$\text{Social cost} = \text{private cost} + \text{External cost}$$

$$\text{Social Benefit} = \text{private Benefit} + \text{External Benefit}$$

* Social cost > private cost,

for e.g. consumption of demerit goods,
The cost to an individual is less than to the whole society, thus there is existence of external cost.



$$MSC > MPC,$$

External cost or Dead weight loss.

3

- ★ The law of diminishing marginal utility is a law of economics stating that as a person increases consumption of a product while keeping consumption of other products constant, there is a decline in the marginal utility that person derives from consuming each additional unit of that product.

4 In the UK in 2015 there were two methods of charging for water supply.

- 1 a fixed charge giving the consumer the right to consume water at zero price per litre
- 2 a price per litre of water used

It has been observed that the amount of water consumed is markedly lower when method 2 is used.

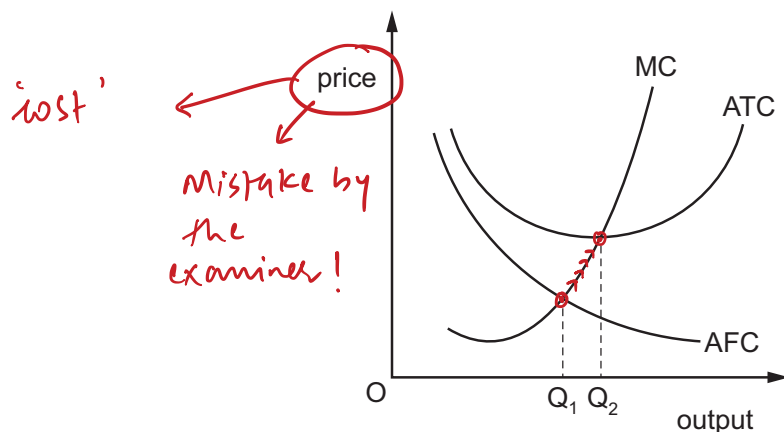
What is implied by this observation?

- A Water in the UK is not a scarce good.
- B Water is a free good of nature and no charge should be levied.
- C Water is a necessity, the use of which should not be restricted by unit prices.

✓ D Water is subject to the law of diminishing marginal utility.

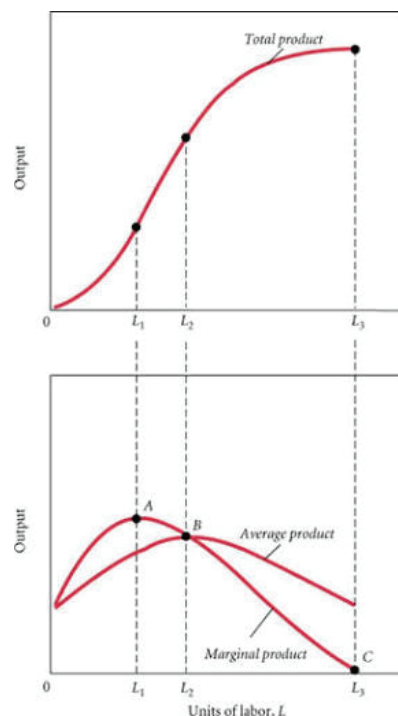
Satisfaction per price i.e. $\frac{MU_{\text{water}}}{P_{\text{water}}}$ is less with every consumption

5 The diagram shows the cost curves for a firm.

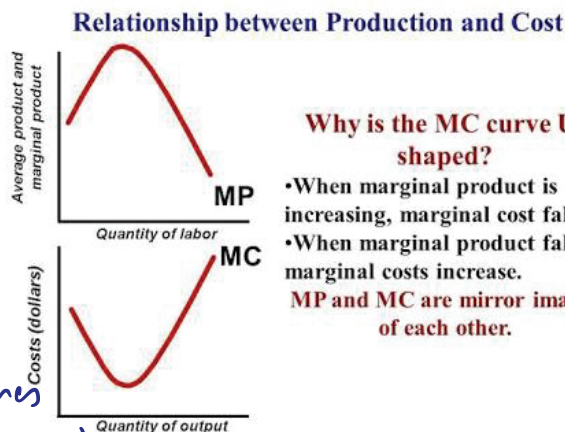


What does the firm experience as it increases output from Q1 to Q2?

- A decreased average variable cost
- ✓ B diminishing returns ($MC \uparrow, MP \downarrow$)
- C economies of scale
- D increased profit



★ As you use more of your variable factors (labour) with your fixed factors (land, capital) of production, total output first increases at an increasing rate ($MP \uparrow$), then increases at a decreasing rate ($MP \downarrow \rightarrow$ Diminishing returns sets in), reaches to maximum ($MP = 0$) and then falls ($MP = -ve$)



Why is the MC curve U-shaped?

- When marginal product is increasing, marginal cost falls.
 - When marginal product falls, marginal costs increase.
- MP and MC are mirror images of each other.**

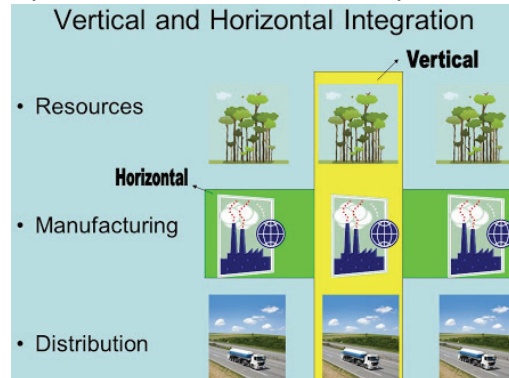
- ★ Horizontal integration is the process of a company increasing production of goods or services at the same part of the supply chain. A company may do this via internal expansion, acquisition or merger. The process can lead to monopoly if a company captures the vast majority of the market for that product or service.

4

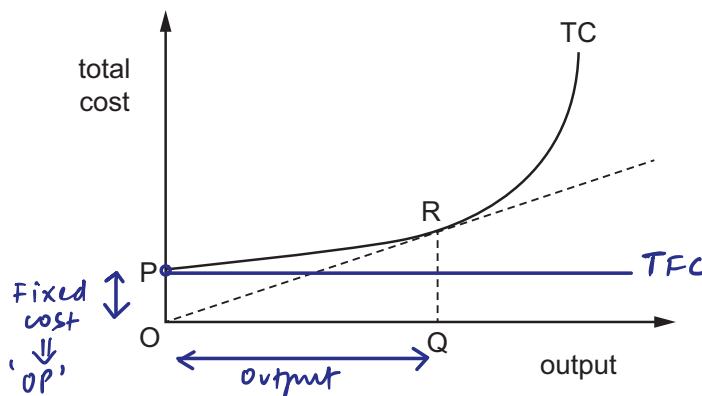
- 6 Firms X and Y merge in a horizontal integration. (same industry, same stage)

What must be true about the industry and the stage of production in which X and Y operate?

	industry	stage of production
A	different	different
B	different	same ✓
C	same ✓	different
D	same ✓	same ✓



- 7 In the diagram the curve TC shows the relationship between a firm's total costs and its level of output.



$$TC = FC + VC$$

At '0' level of output

$$TC = FC + VC$$

$$TC = OP + 0$$

$$TC = OP,$$

$$TC = FC$$

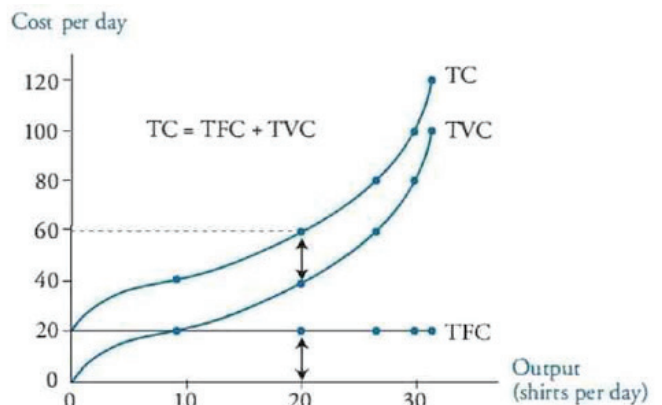
At output OQ average fixed costs are $AFC = \frac{\text{Total fixed cost}}{\text{output}} = \frac{OP}{OQ}$

A equal to $\frac{OP}{QR}$.

B equal to $\frac{QR}{OQ}$.

C equal to $\frac{OP}{OQ}$.

D measured by the slope of the line OR.



★ $\text{Total cost} = \text{Total Fixed cost} + \text{Total variable cost}$

★ Vertical distance between total cost and total variable cost is total fixed cost

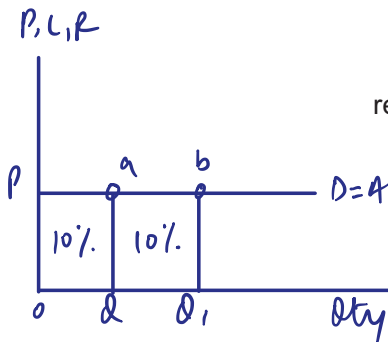
★ Total cost curve starts from fixed cost as at zero level of

output, the only cost paid is total fixed cost

5

Homogeneous
Price taker
perfect knowledge
No Advertising

8 Which diagram shows the total revenue function for a firm in perfect competition?

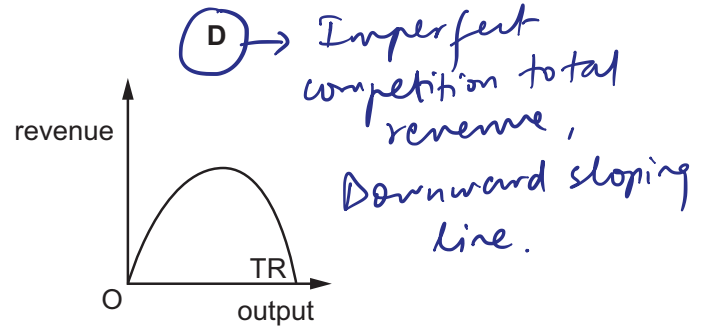
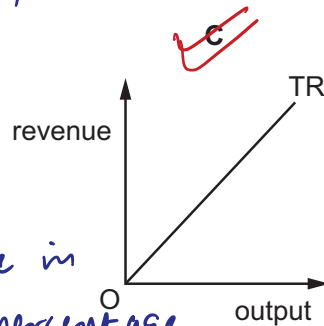
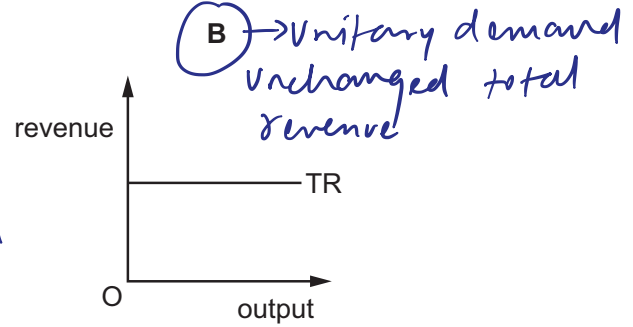
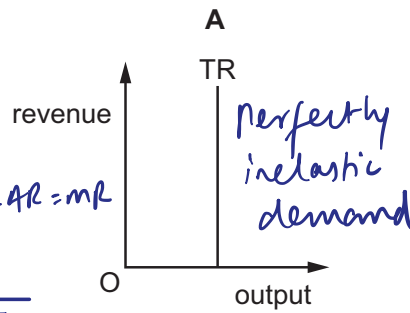


$$TR_{old} = OPaQ$$

$$TR_{new} = OPbQ_1$$

Qty ↑ TR ↑

percentage change in Qty is equal to percentage change in total revenue.



9 Many public utilities can be described as 'natural' monopolies. (see diagram Below) + Examples

Which statement best describes the situation leading to a 'natural' monopoly?

- ☒ A There are high fixed costs and falling average costs over all outputs demanded.
- ☐ B There are legal restrictions on new entrants.
- ☐ C A single firm controls the supply of raw materials.
- ☐ D The firm has a patent on an essential process.

* A natural monopoly is a type of monopoly that exists due to the high fixed or start-up costs of conducting a business in a specific industry. Additionally, natural monopolies can arise in industries that require unique raw materials, technology or similar factors to operate.

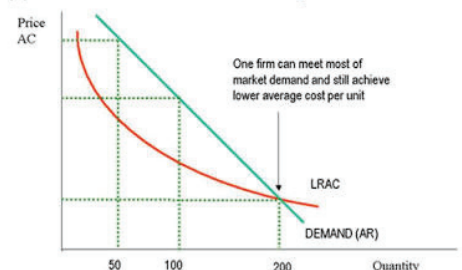
10 Increased advertising by a firm in an imperfectly competitive industry leads to an increase in demand for the industry's product but a fall in the firm's profits.

What could help to explain this?

- ☐ A Production is subject to diseconomies of scale. (Extrant is regarding consumption)
- ☒ B Rival firms respond by increasing their advertising outlays. (Parallel advertising) (Profits ↓)
- ☐ C The demand for the industry's product is price-inelastic. (Profits would have risen)
- ☐ D The increase in demand for the firm's output is entirely at the expense of other firms. (No)

NATURAL MONOPOLY

One unique feature of a natural monopoly (not demonstrated on previous page):
AR (P) intersects LRAC curve when average costs are still **DECLINING**



* Examples of infrastructure include cables and grids for electricity supply, pipelines for gas and water supply, and networks for rail and underground. These costs are also sunk costs, and they deter entry and exit. Railways are often considered a typical example of a natural monopoly.

- 11 An industry consists of a dominant firm, which acts as a price leader, and a large number of small firms.

Which statement about the profit-maximising output of the small firms is correct?

A Average cost is equal to average revenue. (Normal profits)

B Average cost is minimised. (Large firm)

☒ C Marginal cost is equal to price. [marginal cost pricing] (marginal cost, cost of producing additional unit is equal to price i.e. "Allocative efficiency")

D Marginal revenue is zero. (sales maximization) (TR = maximum)

- 12 There are two firms in an industry. Firm X faces a choice. It can either act independently or work with its rival. If it acts independently its profit could be \$900 a week but it could be only \$400 a week depending on what its rival does. If it works with its rival the joint profit of the two firms together would be \$1400, \$700 each. It has no knowledge of what the rival's policy will be.

see Diagram Below

Which concept describes this situation?

A contestable market [A monopoly being a monopoly can't act as a monopoly ----]

B kinked demand curve [Price Rigidity, Fix Price, interdependence]

C principal agent problem [Large firms, share holder and manager conflict]

☒ D prisoner's dilemma [About choice] * The prisoner's dilemma is a standard example of a game analyzed in game theory that shows why two completely rational individuals might not cooperate, even if it appears that it is in their best interests to do so.

↓
Zoi, Zm (Advertising Story)

- 13 What explains the kinked demand curve model of price rigidity in oligopoly? (Diagram Below)

A collusion between all firms in the industry in the setting of prices

B the assumption that a single firm acts as price leader for all firms in the industry

☒ C the individual firm's expectations about other firms' responses to its price changes Elastic and inelastic

D the presence of barriers to the entry of new firms into the industry

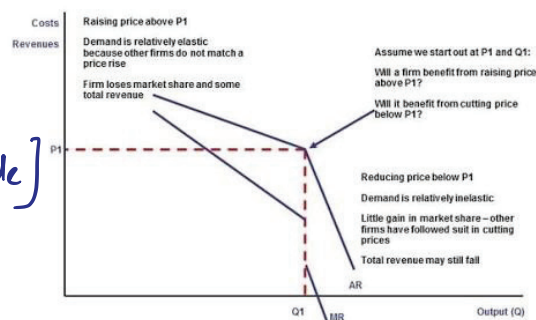
- 14 Which policy is not designed to correct a market failure?

A government provision of health care

☒ B removing import quotas [Moving towards free trade]

C price controls on large firms

D regulations to limit smoke pollution



option A:- corrects the issue of under allocation of merit goods

option C:- corrects the issue of monopoly's abnormal profits

option D:- corrects the issue of over allocation of demerit goods

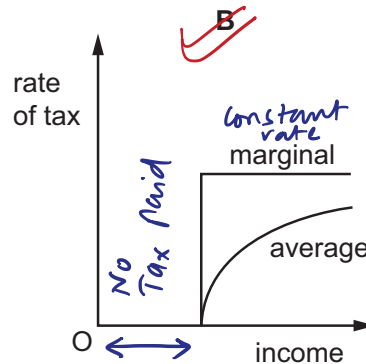
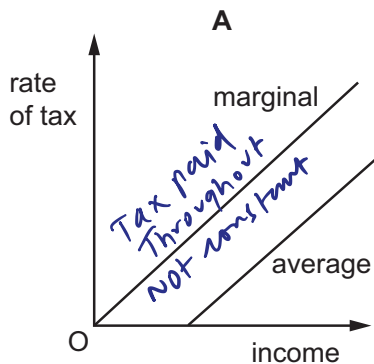
THE PRISONER'S DILEMMA

	B stays silent (cooperates)	B betrays A (defects)
A stays silent (cooperates)	Both serve 1 year	A serves 3 years, B goes free
A betrays B (defects)	A goes free, B serves 3 years	Both serve 2 years

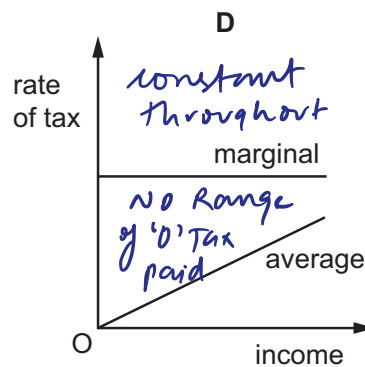
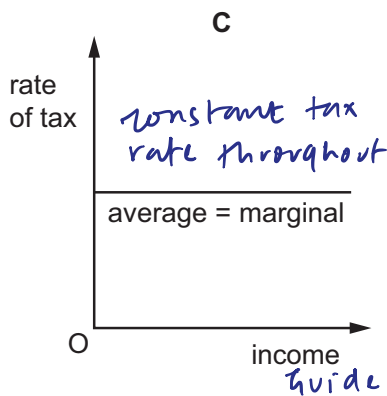
“Only ‘option B’ where ‘0’ rate of tax paid with income earned”

- 15 A tax system allows people to earn a certain sum of income free of income tax. On income above this level, tax is levied at a constant rate.

Which diagram correctly shows how marginal and average rates of tax vary with income?



* The marginal tax rate is the percentage of tax applied to your income for each tax bracket in which you qualify. In essence, the marginal tax rate is the percentage taken from your next dollar of taxable income above a pre-defined income threshold.



* The average tax rate is the total amount of tax divided by total income. For example, if a household has a total income of \$100,000 and pays taxes of \$15,000, the household's average tax rate is 15 percent. The marginal tax rate is the incremental tax paid on incremental income.

- 16 What is the best example of nudge theory as applied to the prevention of tax evasion?

- A employing an extensive administration to ensure detection of evasion
- B imposing heavy penalties to deter offending behaviour
- ☒ C providing information to taxpayers about the undesirable effects of tax evasion
- D requiring employers to inform the tax authorities of workers' pay

* Nudge is a concept in behavioral science, political theory and economics which proposes positive reinforcement and indirect suggestions as ways to influence the behavior and decision making of groups or individuals

- 17 In many countries it has been observed that, when high marginal rates of income tax imposed on richer groups are reduced, the tax revenue raised increases.

What is an implication of this observation? “Option A, B and D are normative”

- A Government policy should aim to minimise the income tax. (No)
- B Sales taxes are always superior to income taxes as a means of raising revenue for the government. (No)
- ☒ C Tax policy intended to reduce income tax on the rich may make the poor better off.
- D The government should always levy proportional rather than progressive income taxes. (No)

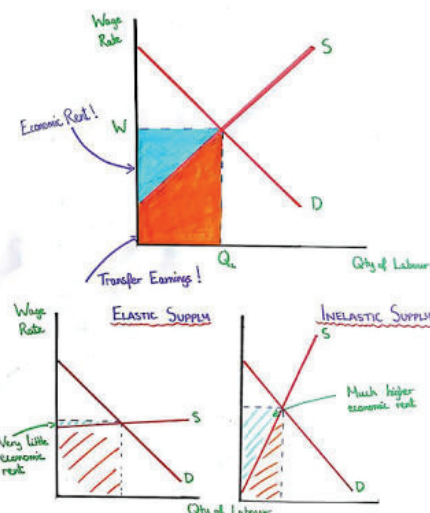
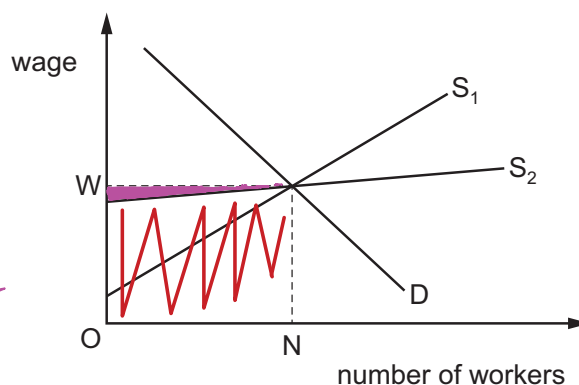
* Reducing the tax on Rich might make them

pay their taxes as high tax to tax evasion, which distorts re-distribution of income.

TRANSFER EARNINGS + ECONOMIC RENT!

18 The diagram represents the market for labour.

W : Transfer earning
 :- Economic Rent
 ↓
 depends on Elasticity of Supply of Labour



What would be the effect on transfer earnings and economic rent of a change in the supply curve from S_1 to S_2 ?

	transfer earnings	economic rent
A	fall	falls ✓
B	fall	rises
C	rise ✓	rises
D	rise ✓	falls ✓

★ Transfer earnings are the minimum reward required to keep factors of production, such as labour, in its current occupation. Diagram showing transfer earnings and economic rent at the market equilibrium wage rate.

★ In economics, economic rent is any payment to an owner or factor of production in excess of the costs needed to bring that factor into production

Rent :- Adjusted with inflation per head :- Takes population

19 How might real GNP per head be adjusted to make it a more reliable indicator when comparing standards of living in different countries?

A adjustments to allow for differences in population size in different countries

B adjustments to allow for differences in rates of inflation in different countries

C adjustments to allow for differences in the relative size of the 'hidden economy' in different countries

D the use of market exchange rates rather than purchasing power parity exchange rates

[It already takes into consideration]

It varies, not reliable

↳ This understates the economy's GDP

20 Which combination of circumstances is most likely to lead to an increase in a country's budget deficit?

Budget deficit :- Government Expenditure > Tax revenue

	total GDP	unemployment	'discretionary' government spending
A	decreases ✓	rises ✓	decreases
B	decreases ✓	rises ✓	no change ✓
C	increases	falls	increases ✓
D	increases	falls	no change

→ 'optional', non-compulsory

* Question refers to "most likely"

Output ↓, Government spending required
 Unemployment ↑, Government spending required

- 21 To combat a rising crime rate, the government decides to recruit additional police officers who are paid a wage equal to that which they previously earned in the private sector.

As long as their previous jobs remain unfilled, what effect will the recruitment of the additional police officers have on the national income and on economic welfare?

	effect on national income	effect on economic welfare
☒ A	increase ✓	unchanged ✓
☐ B	uncertain	decrease
☐ C	unchanged	decrease
☐ D	unchanged	uncertain

Recruitment of additional police officers
↓

Income generated
↓
national income ↑

Previous jobs remain unfilled
↓
unchanged economic welfare

- 22 What could not exist in a closed, mixed economy?

- ☒ A a balance of trade deficit (No Abroad sector)
☐ B a budget deficit (Govt spending > Tax revenue) (private and public sector)
☐ C household debt (Banking sector exists) (government)
☐ D national debt (Government exists)

* The accelerator effect in economics refers to a positive effect on private fixed investment of the growth of the market economy (measured e.g. by a change in Gross Domestic Product).

- 23 What does the accelerator principle state?

- ☐ A Consumption is a function of the rate of change of income. (marginal propensity to consume)
☐ B Income is a function of the rate of change of investment. (multiplier)
☒ C Investment is a function of the rate of change of income. [Accelerator]
☐ D Investment is a function of the rate of interest. (marginal efficiency of capital/investment)

- 24 What would not be classified as structural unemployment? (Fall in demand of particular sector, occupational immobile)

- ☐ A a car worker who is replaced by a robot on the production line (immobile)
☐ B a coal miner whose mine closes because of increased use of solar power (Structural change)
☒ C a fruit picker who cannot find work in the winter months (seasonal)
☐ D a textile worker whose factory closes as production moves abroad (Demand for particular factory ↓)

- 25 What represents a leakage from the circular flow of income in an economy?

- ☐ A a balance of trade surplus ($X(TR) > M(IE)$) (Injections)
☒ B a government budget surplus (Tax revenue > Government spending) (leakage)
☐ C reduction in funds for research and development (funds used elsewhere)
☐ D re-investment of a firm's profits (injections)

26 What will cause a downward shift in the aggregate demand curve of an economy?

- A an increase in firms' stocks of raw materials $\rightarrow$ Aggregate supply $\uparrow$ or Investment $\uparrow = AD \uparrow$
- ☒ B a decrease in the export of goods $[C + I + G + (X - M)] = AD \downarrow$
- C a decrease in the propensity to import $MPM \downarrow, MPC \uparrow, AD \uparrow$
- D a decrease in saving $Saving \downarrow, Consumption \uparrow, AD \uparrow$

27 Which policy is most likely to raise the level of total demand in an economy?

- A an increase in direct taxation matched by a reduction in indirect taxation $\left[\begin{array}{l} \text{Direct tax } \uparrow, \text{ Disposable} \\ \text{income } \downarrow, C \downarrow, AD \downarrow \end{array} \right]$
- B a relaxation of import controls $M(TE) \uparrow, AD \downarrow$
- C cuts in public expenditure matched by tax cuts $(D.Y \uparrow, C \uparrow, AD \uparrow)$ $\left[\begin{array}{l} \text{Indirect tax } \downarrow, S \uparrow, \\ \text{prices } \downarrow, C \uparrow, AD \uparrow \end{array} \right]$
- ☒ D devaluation of the currency $\left[\begin{array}{l} X \text{ prices } \downarrow, X(TE) \uparrow \\ M \text{ prices } \uparrow, M(TE) \downarrow \end{array} \right] \rightarrow AD \uparrow$

28 The government cuts taxes and increases its spending in order to create jobs during a recession.

When will a resulting budget deficit be least likely to cause a permanent reduction in unemployment?

- ☒ A when households expect higher inflation rates in the future $\left[\begin{array}{l} \text{with spending, they will be} \\ \text{saving too} \end{array} \right]$
- B when households spend their extra disposable incomes on domestic products (Permanent reduction)
- C when the government's extra spending is on new infra-structure projects (Permanent reduction)
- D when the government's extra spending is on the salaries of public sector workers (Permanent reduction)

This restricts growth!

29 In 2011, the global money supply, narrowly defined, rose 11% and broadly defined money increased roughly 8%, as central banks continued efforts to keep interest rates low.

What type of policy does this statement illustrate?

- A contractionary fiscal $(Tax \uparrow, G \downarrow)$ (Not mentioned)
- B contractionary monetary $(M^s \downarrow, R \uparrow)$ (Not mentioned)
- C expansionary fiscal $(Tax \downarrow, G \uparrow)$ (Not mentioned)
- ☒ D expansionary monetary $M^s \uparrow, R \downarrow, C \uparrow, I \uparrow, AD \uparrow$

30 Which policy measure would be most likely to reduce a developing economy's birth rate and its household saving ratio?

- A an increase in student tuition fees (No link to Birth rate)
- B a relaxation of immigration controls (No link to Birth rate)
- ☒ C the introduction of a basic state pension $[They \text{ will earn even when not working}]$
- D the removal of tariffs on imported foodstuffs (No link to birth rate)

This can reduce family size as mostly due to earning issue, larger families exists!